

Vishal Ramesh

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PROFESSIONAL SUMMARY

Results-driven Machine Learning Engineer with over 5 years of experience designing, building, and deploying scalable end-to-end ML solutions in production cloud environments. Expertise in developing production-grade systems that handle millions of requests daily, including advanced NLP pipelines, decision support systems, and robust large-scale model deployment infrastructures. Specializes in microservice architecture (Java, Spring Boot) and MLOps best practices, effectively translating complex enterprise challenges into impactful, data-driven applications.

WORK EXPERIENCE

Lead Data Scientist

July 2025 - Present

Freshworks, Chennai, India

- Directed transition of features to open-source LLaMA and Qwen models, utilizing quantization techniques and assessing output quality with MLflow's LLM-as-a-Judge framework.
- Spearheading enhancements to the RCA product feature to support higher customer data volumes and improved anomaly detection throughput.
- Architecting an organization-wide evaluation framework for agent-based AI features, standardizing quality measurement across agentic workflows.

Senior Data Scientist

July 2022 - June 2025

Freshworks, Chennai, India

- Engineered and implemented a production-scale Custom Intent Detection microservice, enabling customers to train models on their own data, overseeing 10,000+ live custom models handling approximately 1 million daily requests.
- Designed and implemented an organization-wide MLOps framework covering model monitoring, feedback loops, and data and model version tracking to enhance operational efficiency.
- Built an Entity Extraction service to identify and extract structured entities from support tickets and live chat conversations.
- Designed and implemented a Query-to-Graph agentic system, enabling customers to build analytical dashboards and conduct data-driven conversations using natural language queries.
- Developed and deployed a Root Cause Analysis (RCA) product feature leveraging Generative AI and data analysis techniques to proactively alert customer business leaders about anomalies in their key business metrics, along with AI-generated potential root causes.

Data Scientist

April 2021 - June 2022

Freshworks, Chennai, India

- Developed and integrated a conversational AI model for small talk interactions within the chatbot framework, improving user experience and dialogue effectiveness.
- Designed and deployed a production-grade Embedding service by quantizing the LaBSE model, building a high-performance inference layer, and deploying it for large-scale production use.

Graduate Trainee - Data Science

October 2020 - March 2021

Freshworks, Chennai, India

- Implemented a Question-and-Answer (QnA) retrieval system and integrated it with the existing AI chatbot to improve response accuracy and coverage.

Data Science Intern

December 2019 - September 2020

Freshworks, Chennai, India

- Enhanced the AI chatbot to incorporate communication channel-based context, enabling more relevant and personalized bot responses across different messaging channels.
- Implemented Global Feedback Flows for the chatbot platform, enabling structured collection of user feedback to drive continuous improvement.

PATENTS

Processing User Queries for a Virtual Agent | US20240354497A1 | Freshworks Inc.

Root Cause Analysis Using Artificial Intelligence Insights | 19/415,304 | Freshworks Inc.

SKILLS

Programming Languages: Python, Java

Frameworks & Libraries: FastAPI, TensorFlow, PyTorch, Llama.cpp

Infrastructure & DevOps: Docker, Redis, AWS, Elasticsearch, CI/CD (Jenkins, AWS CodePipeline)

ML & AI: LLM Evaluation, MLflow, NLP, Intent Detection, Semantic Parsing, Entity Extraction, Model Deployment, MLOps

EDUCATION

Bachelor of Technology in Computer Science and Engineering

January 2020

SASTRA Deemed to be University, Thanjavur, India

Relevant Coursework: Data Structures, Database Management Systems, Artificial Intelligence, Web Development, Object-Oriented Design